

Statement of Research Interests

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PhD Position: Structural Generalization in Transformer-based LLMs
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Research Motivation

My research interests lie at the intersection of structural inductive biases and the generalization capabilities of deep sequence models. During my engineering degree at ESI-SBA, I have been fascinated by how different architectural constraints (Transformers vs. State Space Models vs. GNNs) influence a model's ability to generalize to unseen structural patterns—a challenge known as "Structural Generalization."

Current Research & Findings

In my current project at Sonatrach, I am investigating the use of **Mamba-SSM (State Space Models)** and **xLSTM** as generative surrogates for industrial digital twins. My key finding is that while Transformers excel at capturing global context, they often struggle with the "Structural Latency Paradox" in real-time CPS environments. By implementing a dual-track architecture (Cloud/Edge), I explored how physics-informed constraints (**PINNs**) can act as a structural bias to ensure thermodynamic consistency in out-of-distribution scenarios.

Furthermore, my work on **Graph Neural Networks (GNNs)** for the Career Connect project allowed me to explore structural generalization in a non-Euclidean space. I researched how network centrality measures can be used to improve node representation stability, which I believe is directly applicable to the "Structural Generalization" problem in LLMs, where the model must handle recursive or long-range dependencies that follow specific structural rules.

Future Goals at Saarland University

The PhD position at Saarland University offers a unique opportunity to formalize these empirical observations into a rigorous theoretical framework. I am particularly interested in:

1. **Structural Inductive Biases:** Investigating how attention mechanisms can be modified to explicitly respect structural priors (e.g., hierarchy, recursion).
2. **Generalization beyond Sequence Length:** Studying how Transformer-based LLMs can be made to generalize to structural complexities that exceed their training window.
3. **Hybrid Architectures:** Exploring the synergy between Transformer-based LLMs and structured representations (like GNNs or SSMs) to enhance robustness.

Conclusion

I am eager to bring my background in AI engineering and my passion for structural modeling to the Saarland University research team. I am confident that my combination of algorithmic understanding and deep learning expertise will allow me to contribute meaningfully to the project.

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